Planning a National Integrated Data Service for Environmental-Justice Research (EJ-IDSS)

Submitted to the U.S. National Science Foundation
Program Solicitation NSF 26-509 — Integrated Data Systems & Services (IDSS)
Category III: Planning Grant | Requested: $499,500 | Duration: 24 months

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Project Summary

Overview. This two-year planning project will define the science requirements, federated technical architecture, and governance model for a national-scale Integrated Data Service for Environmental-Justice research (EJ-IDSS). Environmental-justice science requires linking environmental-exposure, public-health, and socioeconomic data that today live in separate, incompatible repositories maintained by federal agencies, states, universities, and community organizations. No national-scale, operations-quality cyberinfrastructure currently federates these data for the environmental-justice research community. Through structured stakeholder engagement, a landscape analysis of existing repositories, and a focused technical prototyping effort, the project will produce a reference architecture, a cross-domain metadata schema, a data-governance and community-consent framework, and an operations and sustainability plan that together prepare a future full-scale IDSS deployment. The effort will federate—not duplicate—existing repositories and will align with NSF's prior cyberinfrastructure investments (CSSI, CC*).

Intellectual Merit. The project advances data cyberinfrastructure by developing shared, reusable standards for linking heterogeneous environmental-exposure and community-health data at national scale—a transdisciplinary problem that no single existing repository solves. The planning effort produces three transferable artifacts: a reference architecture for federating geographically and administratively distributed datasets; a cross-domain metadata schema that maps exposure, health, and socioeconomic vocabularies to community standards (DataCite, schema.org/Dataset, OGC); and a governance model for sensitive, community-contributed data.

Broader Impacts. Led by Morgan State University, a historically Black university, the project trains graduate and undergraduate students from groups underrepresented in computing and data science in modern data cyberinfrastructure and community-engaged research. It broadens participation by partnering with community organizations in overburdened neighborhoods so residents help shape the data service rather than serving as passive subjects, and by opening its planning workshops to regional minority-serving institutions. A formal evaluation plan, led by an external evaluator, will measure training outcomes, partner engagement, and the usability of the proposed architecture against pre-defined targets.

Project Description

1. Vision, Goals, and Driving Requirements

Our vision is a national, operations-quality data service that lets any researcher discover, access, and responsibly combine environmental-exposure, health, and socioeconomic data to study environmental injustice and its remedies. This planning project does not build that operational system; it produces the validated requirements, architecture, and plan needed to propose one. We pursue three concrete goals: (G1) define the driving science requirements with researchers and affected communities; (G2) specify a federated technical architecture and concept of operations; and (G3) produce an operations, governance, and sustainability plan suitable for a future Category I or II IDSS proposal. The driving requirements are explicit and measurable: the future service must federate at least five major external repositories without duplicating them; return cross-domain queries over exposure and health data in under five seconds for common requests; and enforce community-consent and data-sovereignty rules at the dataset level.

2. Background and Significance

Environmental-justice research is constrained today because the data it depends on are fragmented. Air- and water-quality monitoring data sit in federal and state environmental systems; health outcomes sit in public-health repositories with strict access controls; and the socioeconomic context lives in yet other census and administrative datasets. Each uses different identifiers, geographies, time scales, and metadata, and linking them today is slow, manual, and rarely reproducible. Prior NSF investments (CSSI, CC*) built valuable capacity at campus and regional scales, but no national-scale, operations-quality service federates these data for the environmental-justice community. Closing this gap is significant: credible findings depend on linking exposure to outcomes at the neighborhood scale, and a shared national service would make that analysis reproducible and auditable.

3. Approach and Planning Activities

We will execute four interlocking activities. **(A) Stakeholder requirements.** Three structured workshops—researchers, data-provider agencies, and community organizations—using a documented elicitation protocol to produce a prioritized requirements register. **(B) Landscape analysis.** Inventory the major candidate repositories, documenting access models, metadata, identifiers, licensing, and governance constraints. **(C) Technical prototyping.** A narrow proof-of-concept that maps and links two representative datasets (an exposure dataset and a health-outcomes dataset) through a candidate metadata schema, to test feasibility—not to deploy a service. **(D) Governance design.** A data-governance and community-consent framework for sensitive and community-contributed data.

4. Evaluation and Expected Outcomes

Success is defined by measurable deliverables and independent evaluation: a prioritized requirements register validated by at least 20 stakeholders; a landscape report covering at least eight repositories; a working prototype that links the two sample datasets and answers three benchmark cross-domain queries; a reference architecture and concept of operations; and a governance framework reviewed by community partners. An external evaluator assesses each deliverable against these targets and measures student-training and partner-engagement outcomes, reporting formatively at the midpoint and summatively at the end.

5. Project Management

The PI (Computer Science) leads coordination and the technical prototype; a co-PI in Public Health leads health-data requirements and governance; and a community-partnerships coordinator leads the workshops. The 24-month timeline has four milestones: **M1 (1–6)** workshops complete and requirements register drafted; **M2 (7–12)** landscape analysis and metadata schema drafted; **M3 (13–18)** prototype linking two datasets demonstrated; **M4 (19–24)** reference architecture, governance framework, and sustainability plan delivered.

References Cited

[1] National Science Foundation. *Proposal & Award Policies & Procedures Guide (PAPPG)*. Current edition.

[2] National Academies of Sciences, Engineering, and Medicine. *Reproducibility and Replicability in Science*. The National Academies Press, 2019.

[3] U.S. EPA. *EJScreen: Environmental Justice Screening and Mapping Tool*. Technical documentation.

[4] Wilkinson, M. et al. “The FAIR Guiding Principles for scientific data management and stewardship.” *Scientific Data* 3, 160018 (2016).

Data Management and Sharing Plan

Types of data. Requirements registers, a repository landscape inventory, a candidate metadata schema, de-identified workshop notes, prototype code, and design documents. The prototype links two small, already-public sample datasets; the project collects no new human-subjects data.

Standards and formats. Open formats (PDF/A, CSV, JSON); metadata follows DataCite and schema.org/Dataset, with geospatial metadata following OGC conventions.

Storage and security. Artifacts are stored on Morgan State's managed research storage with automated backup; sensitive material is de-identified and access-controlled.

Sharing and access. Reports, the metadata schema, and prototype code are deposited in a public repository (Zenodo) under open licenses (CC-BY for documents, Apache-2.0 for code) with DOIs at each milestone, not merely “available on request.”

Retention. Shared artifacts are retained for at least five years after the award, consistent with NSF policy and Morgan State's records schedule.

Budget Justification (Summary)

Senior personnel. PI (Computer Science) and co-PI (Public Health) at partial academic-year and summer effort. **Other personnel.** Two graduate research assistants support the landscape analysis, prototype, and workshop logistics. **Fringe** is applied at Morgan State's negotiated rates. **Travel** supports three stakeholder workshops and one NSF PI meeting. **Materials and other direct costs** cover modest cloud prototyping resources, participant-support and workshop costs, and external evaluation. **Indirect costs (F&A)** are applied to modified total direct costs at Morgan State's federally negotiated rate. The total request remains within the $500,000 Category III ceiling; voluntary committed cost sharing is not included, as required by the solicitation.